

Quick Activity 1: School Supplies Calculator

1. Problem Analysis

The program will calculate the total cost of school supplies by multiplying the quantity of each item by its unit price, then adding all the subtotals.

The program should determine:

- The cost of each school supply.
- The total amount to be paid.

2. Inputs–Process–Outputs

Input	Process	Output
Quantity and price of notebooks	Multiply quantity x price	Notebook subtotal
Quantity and price of pens	Multiply quantity x price	Pen subtotal
Quantity and price of pencils	Multiply quantity x price	Pencil subtotal
Quantity and price of erasers	Multiply quantity x price	Eraser subtotal
All subtotals	Add all subtotals	Total cost

Formula:

Subtotal = Quantity × Unit Price

Total Cost = Sum of all Subtotals

3. Decomposition

The problem can be divided into these smaller tasks:

1. Get the quantity and unit price of each school supply.
2. Calculate the subtotal for each item.
3. Add all the subtotals.
4. Display the total cost.

4. Pseudocode

START

INPUT notebook_quantity
INPUT notebook_price

INPUT pen_quantity
INPUT pen_price

INPUT pencil_quantity
INPUT pencil_price

INPUT eraser_quantity
INPUT eraser_price

notebook_total = notebook_quantity × notebook_price
pen_total = pen_quantity × pen_price
pencil_total = pencil_quantity × pencil_price
eraser_total = eraser_quantity × eraser_price

total_cost = notebook_total + pen_total + pencil_total + eraser_total

DISPLAY notebook_total
DISPLAY pen_total
DISPLAY pencil_total
DISPLAY eraser_total
DISPLAY total_cost

END

5. Dry Run — Test Case 1

Given:

- 3 notebooks $\times$ ₱40 = ₱120
- 5 pens $\times$ ₱15 = ₱75
- 4 pencils $\times$ ₱10 = ₱40
- 2 erasers $\times$ ₱8 = ₱16

Calculation:

Item	Quantity	Unit Price	Subtotal
Notebook	3	₱40	₱120
Pen	5	₱15	₱75
Pencil	4	₱10	₱40
Eraser	2	₱8	₱16

Total Cost = ₱120 + ₱75 + ₱40 + ₱16

Total Cost = ₱251

Output:

Total cost: ₱251

6. Dry Run — Test Case 2

Given:

- 5 notebooks $\times$ ₱40 = ₱200
- 8 pens $\times$ ₱15 = ₱120
- 6 pencils $\times$ ₱10 = ₱60
- 3 erasers $\times$ ₱8 = ₱24

Calculation:

Item	Quantity	Unit Price	Subtotal
Notebook	5	₱40	₱200
Pen	8	₱15	₱120
Pencil	6	₱10	₱60
Eraser	3	₱8	₱24

Total Cost = ₱200 + ₱120 + ₱60 + ₱24

Total Cost = ₱404

Output:

Total cost: ₱404